

MANIPAL UNIVERSITY JAIPUR
Department of Artificial Intelligence and Machine Learning (AIML)

Course Hand-out

A. Basic Details

Programme Name:	B.Tech
Course Name:	NoSQL Database
Course Code:	AIM3141
LTPC (Lecture Tutorial Practical Credits):	3 Credits 3 0 0 3
Session:	July 2026 – December 2026
Class:	5th Semester
Course Coordinator:	Dr. Arunangshu Pal
Course Instructor(s):	Dr. Arunangshu Pal, Dr. Rahul Sharma, Ms Neny Pandel, Ms. Surabhi Agrawal, Mr. Pramod Kumar
Additional Practitioner(s) – if any (Industry Fellow / Visiting Faculty / Adjunct Faculty, etc.):	—

B. Introduction

The NoSQL Database course introduces modern data-management technologies designed for large-scale, distributed, semi-structured, unstructured, and AI-oriented workloads. The course begins with the limitations of traditional relational systems and develops a foundation in document, key-value, column-family, graph, and vector databases. Students study distributed-system principles such as ACID and BASE properties, CAP and PACELC theorems, eventual consistency, partitioning, sharding, replication, and consistent hashing. MongoDB is used as the principal platform for data modelling, CRUD operations, indexing, aggregation, schema validation, application connectivity, change streams, security, cloud deployment, and performance optimization. The course further introduces vector search, similarity metrics, approximate nearest-neighbour methods, and a simple retrieval-augmented generation back-end. Through demonstrations, case studies, and a hands-on application, students develop the ability to choose and use suitable NoSQL technologies for contemporary web, cloud, IoT, analytics, and artificial-intelligence applications.

C. Course Outcomes

After successful completion of the course, the learner will be able to achieve the following outcomes. All Course Outcomes are restricted to Revised Bloom's Taxonomy cognitive levels 2–4.

CO Statement	CO	Bloom's Level	Target Attainment %	Target Attainment Level
Explain the principles of NoSQL and distributed data systems, including database types, ACID/BASE, CAP, PACELC, eventual consistency, and partitioning strategies.	AIM3141.1	Understanding (L2)	80%	2.40
Apply MongoDB CRUD, querying, indexing, and schema-design techniques to structured and semi-structured datasets.	AIM3141.2	Applying (L3)	80%	2.40
Develop MongoDB aggregation pipelines, schema validation, application connectivity, and change streams for data-driven applications.	AIM3141.3	Applying (L3)	80%	2.40
Analyze MongoDB architectures for sharding, replication, security, cloud deployment, backup, and performance requirements.	AIM3141.4	Analyzing (L4)	80%	2.40
Analyze the performance and suitability of Redis, Neo4j, and vector-database solutions for similarity search, AI, and RAG-oriented workflows.	AIM3141.5	Analyzing (L4)	80%	2.40

Note: The Bloom's level indicates the cognitive level of Revised Bloom's Taxonomy.

Information about attainment levels:

Attainment (%)	Level
< 60%	0
≥ 60% and < 70%	1
≥ 70% and < 80%	2
≥ 80%	3

D. Program Outcomes and Program Specific Outcomes

Program Specific Outcomes (PSOs)

PSO1: Graduates will be able to examine the applications of Artificial Intelligence and Machine Learning in real-life problems.

PSO2: Graduates will be able to design and implement intelligent systems for multidisciplinary problems.

Program Outcomes (POs)

- 1. Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- 2. Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- 3. Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.
- 4. Conduct investigations of complex problems:** Use research-based knowledge and research methods, including design of experiments, analysis and interpretation of data, and synthesis of information, to provide valid conclusions.
- 5. Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering activities with an understanding of their limitations.
- 6. The engineer and society:** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to professional engineering practice.
- 7. Environment and sustainability:** Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate knowledge of and need for sustainable development.
- 8. Ethics:** Apply ethical principles and commit to professional ethics, responsibilities, and norms of engineering practice.
- 9. Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams and multidisciplinary settings.
- 10. Communication:** Communicate effectively on complex engineering activities with the engineering community and society at large through effective reports, design documentation, presentations, and clear instructions.
- 11. Project management and finance:** Demonstrate knowledge and understanding of engineering and management principles and apply them to one's own work, as a member and leader in a team, to manage projects in multidisciplinary environments.
- 12. Life-long learning:** Recognize the need for, and have the preparation and ability to engage in, independent and life-long learning in the broadest context of technological change.

E. Assessment Plan

Criteria	Description	Maximum Marks
Internal Assessment (Summative)	Mid-Term Examination (closed book) [Approximately 60% of the syllabus]	30
Internal Assessment (Summative)	Quiz (2): 20 marks Research Project/Mini Project: 10 marks	30
End-Term Examination (Summative)	End-Term Examination (closed book)	40
	Total	100

Attendance (Formative)

A minimum of 75% attendance is required for a student to be eligible to appear in the End-Term Examination. The permitted 25% absence includes all categories of leave, including medical leave.

F. Syllabus

Module 1: Foundations of NoSQL and Distributed Systems: Overview of traditional RDBMS limitations and the emergence of NoSQL. Types of NoSQL databases: Document, Key-Value, Column-Family, and Graph. Key concepts: ACID vs. BASE, CAP Theorem, and Eventual Consistency. The PACELC Theorem: trade-off between latency and consistency during normal operation. Data partitioning strategies: horizontal vs. vertical sharding and consistent hashing. Use cases and industry motivation.

Module 2: MongoDB – Architecture and Basic Operations: Introduction to MongoDB, installation, and environment setup using Shell, Compass, and Atlas. Data model: documents, collections, and BSON types. CRUD operations, projections, and querying embedded documents. Indexing, filtering, and sorting. Advanced schema design using the Outlier Pattern, Computed Pattern, and Extended Reference Pattern.

Module 3: Aggregation and Application Development: Aggregation pipeline, stages, and operators. Data modelling: embedded vs. referenced documents and schema validation. Introduction to application development using PyMongo (Python) or Node.js. Reactive data streams using MongoDB Change Streams for real-time application updates. Hands-on project: building a CRUD-based application.

Module 4: Advanced Concepts and Cloud Ecosystem: Data partitioning, sharding, and replica sets for high availability. MongoDB security: user roles, authentication, and encryption at rest. Backup, restore, monitoring, and performance optimization. Introduction to Database-as-a-Service (DBaaS): managing NoSQL in the cloud using AWS DynamoDB and MongoDB Atlas.

Module 5: Vector Databases and Polyglot Case Studies: Introduction to vector databases for storing and querying high-dimensional embeddings. Vector search fundamentals: similarity metrics (cosine and Euclidean) and approximate nearest-neighbour algorithms. Overview of Redis for caching and Neo4j for graph-based recommendations. Hands-on AI case study: building a simple retrieval-augmented generation back-end using a vector store such as Pinecone or Milvus. Comparative matrix and guidelines for selecting a NoSQL solution.

Reference Books and Learning Resources

1. T. Dan Sullivan, NoSQL for Mere Mortals, Pearson Education, 2015.
2. Kristina Chodorow, MongoDB: The Definitive Guide, O'Reilly Media.
3. Martin Kleppmann, Designing Data-Intensive Applications, O'Reilly Media.
4. Eric Redmond and Jim R. Wilson, Seven Databases in Seven Weeks, O'Reilly Media.
5. MongoDB Documentation, AWS DynamoDB Documentation, and official documentation of Pinecone, Milvus, and Weaviate for vector search.

G. Lecture Plan

The lecture plan indicates the schedule of content delivery, measurable session outcomes, corresponding Course Outcomes, mode of delivery, and mode of assessment.

SL. No.	Topic	Session Outcome	Corresponding CO	Mode of Delivery	Mode of Assessing CO
1	Course orientation; limitations of traditional RDBMS	Explain the motivation for NoSQL systems.	AIM3141.1	Interactive lecture	Quiz, MTE, ETE
2	Evolution and industry need for NoSQL	Explain drivers for scalable and flexible data stores.	AIM3141.1	Lecture + discussion	Quiz, MTE, ETE
3	NoSQL types: document, key-value, column-family, graph	Classify NoSQL database families and use cases.	AIM3141.1	Lecture + examples	Quiz, MTE, ETE
4	ACID versus BASE properties	Differentiate ACID and BASE transaction models.	AIM3141.1	Concept mapping	Quiz, MTE, ETE
5	CAP theorem and eventual consistency	Explain CAP trade-offs and eventual consistency.	AIM3141.1	Lecture + case analysis	Quiz, MTE, ETE
6	PACELC theorem	Interpret latency-consistency trade-offs under PACELC.	AIM3141.1	Inquiry-based learning	Quiz, MTE, ETE
7	Partitioning: horizontal, vertical, consistent hashing; use cases	Compare partitioning strategies for distributed workloads.	AIM3141.1	Problem-based learning	Quiz, MTE, ETE
8	MongoDB overview, installation, Shell, Compass, and Atlas	Configure a MongoDB working environment.	AIM3141.2	Demonstration + hands-on	Quiz, MTE, ETE
9	MongoDB architecture; documents, collections, and BSON	Explain MongoDB architecture and BSON data types.	AIM3141.2	Lecture + demonstration	Quiz, MTE, ETE
10	CRUD operations: insert and find	Apply insert and retrieval operations.	AIM3141.2	Lecture + Guided practice	Lab task, MTE, ETE
11	CRUD operations: update and delete	Apply update and delete operations with suitable filters.	AIM3141.2	Lecture + Guided practice	Lab task, MTE, ETE
12	Projections and querying embedded documents	Construct queries for nested and embedded documents.	AIM3141.2	Problem solving session	Quiz, MTE, ETE
13	Filtering, sorting, and indexing	Apply filtering, sorting, and index creation.	AIM3141.2	Lecture + Demonstration	Assignment, MTE, ETE
14	Schema design principles; Outlier Pattern	Apply the Outlier Pattern to irregular data distributions.	AIM3141.2	Case based learning	Assignment, MTE, ETE
15	Computed Pattern and Extended Reference Pattern	Apply advanced MongoDB schema patterns.	AIM3141.2	Case based learning	Assignment, MTE, ETE
16	Aggregation framework: pipeline and stages	Implement a basic aggregation pipeline.	AIM3141.3	Lecture + demonstration	Quiz, MTE, ETE
17	Aggregation operators: match, project, group, sort	Implement common aggregation operators.	AIM3141.3	Case based learning + Guided practice	Lab task, MTE, ETE
18	Advanced aggregation: lookup, unwind, facet	Construct multi-stage analytical pipelines.	AIM3141.3	Problem based learning	Assignment, MTE, ETE
19	Embedded versus referenced documents	Analyze modelling choices for application requirements.	AIM3141.3	Case discussion	Quiz, MTE, ETE
20	Schema validation and data integrity	Implement validation rules for MongoDB collections.	AIM3141.3	Case based learning + practice	Lab task, MTE, ETE
21	Application connectivity using PyMongo	Connect a Python application to MongoDB.	AIM3141.3	Case based learning + Hands on	Project, MTE, ETE
22	Application connectivity using Node.js	Connect a Node.js application to MongoDB.	AIM3141.3	Hands-on coding	Project, Quiz, ETE
23	MongoDB Change Streams and reactive updates	Implement real-time event handling with Change Streams.	AIM3141.3	Case based learning + Hands on	Project, Quiz, ETE
24	CRUD application: planning and schema design	Prepare the architecture and schema of a CRUD application.	AIM3141.3	Project based learning	Project review
25	CRUD application: implementation and testing	Implement and test the CRUD application.	AIM3141.3	Project based learning	Project evaluation
26	Sharding architecture and shard-key selection	Analyze sharding choices for scalable workloads.	AIM3141.4	Lecture + case analysis	Quiz, ETE

SL. No.	Topic	Session Outcome	Corresponding CO	Mode of Delivery	Mode of Assessing CO
27	Replica sets, failover, and high availability	Analyze replication and failover behaviour.	AIM3141.4	Simulation + discussion	Quiz, ETE
28	Security: users, roles, authentication, encryption at rest	Analyze security controls for a MongoDB deployment.	AIM3141.4	Lecture + demonstration	Quiz, ETE
29	Backup, restore, monitoring, and performance optimization	Analyze operational strategies for reliability and performance.	AIM3141.4	Lecture + Case based learning	Assignment, ETE
30	DBaaS: MongoDB Atlas and AWS DynamoDB	Compare managed NoSQL services and deployment considerations.	AIM3141.4	Lecture	Certification, ETE
31	Vector databases and high-dimensional embeddings	Explain the role of vector databases in AI systems.	AIM3141.5	Lecture + visualization	Quiz, ETE
32	Similarity metrics: cosine and Euclidean distance	Analyze similarity measures for vector retrieval.	AIM3141.5	Lecture	Quiz, ETE
33	Approximate nearest-neighbour search	Analyze ANN methods and retrieval trade-offs.	AIM3141.5	Lecture + demonstration	Assignment, ETE
34	Vector stores: Pinecone, Milvus, and Weaviate	Compare vector-store capabilities and deployment options.	AIM3141.5	Comparative case study	Group task, ETE
35	Redis caching and Neo4j graph recommendations	Analyze caching and graph-database use cases.	AIM3141.5	Peer learning + case study	Group task, ETE
36	Simple RAG back-end; comparative matrix; project presentations	Analyze and justify a suitable NoSQL solution for an AI workflow.	AIM3141.5	Project presentation	Project, ETE

H. Learner-Centric Activities

Low Difficulty Level Activities:

Guided peer-learning sessions on NoSQL database families, real-world use cases, and terminology. Short hands-on practice using MongoDB Shell, Compass, or Atlas for CRUD operations, filters, and basic aggregation. Curated reading and video-based exploration of ACID/BASE, CAP, PACELC, and eventual consistency.

High Difficulty Level Activities:

Case-based exercises on schema design, indexing strategy, validation rules, and aggregation pipeline construction.

Mini project involving a MongoDB-backed application with application connectivity, change streams, and reporting.

Comparative investigation of Redis, Neo4j, and a vector database for AI-driven search, recommendation, or RAG scenarios.

Independent exploration of sharding, replication, security controls, backup, and cloud deployment through guided labs or official documentation.

I. Outside Class Engagement Activities

Activity	Difficulty Level	Hours Engaged
Participation in a webinar, guest lecture, or technology talk related to MongoDB, distributed databases, graph systems, or vector search.	Medium	2 hrs
Group study and peer-review sessions focused on querying, aggregation pipelines, schema design, and indexing decisions.	Medium	2 hrs
Development of a small NoSQL application, dashboard, or case study using MongoDB with PyMongo/Node.js or Atlas.	High	8 hrs
Completion of a guided online lab, tutorial series, or certification module on MongoDB Atlas, Redis, Neo4j, or vector databases.	High	10 hrs

J. Course Articulation Matrix

Correlation levels: 1 = Low correlation, 2 = Moderate correlation, 3 = Substantial correlation; blank = no direct correlation.

CO	Statement	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
1	Explain the principles of NoSQL and distributed data systems, including database types, ACID/BASE, CAP, PACE/LC, eventual consistency, and partitioning strategies.	3	2			1							1	1	
2	Apply MongoDB CRUD, querying, indexing, and schema-design techniques to structured and semi-structured datasets.	3	2	2		3				1			1	2	2
3	Develop MongoDB aggregation pipelines, schema validation, application connectivity, and change streams for data-driven applications.	2	2	3	1	3				2	1	1	2	2	3
4	Analyze MongoDB architectures for sharding, replication, security, cloud deployment, backup, and performance requirements.	2	3	2	2	2	1		2				2	1	2
5	Analyze the performance and suitability of Redis, Neo4j, and vector-database solutions for similarity search, AI, and RAG-oriented workflows.	2	3	3	2	3				2	2		3	3	3

Aruna Singh

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